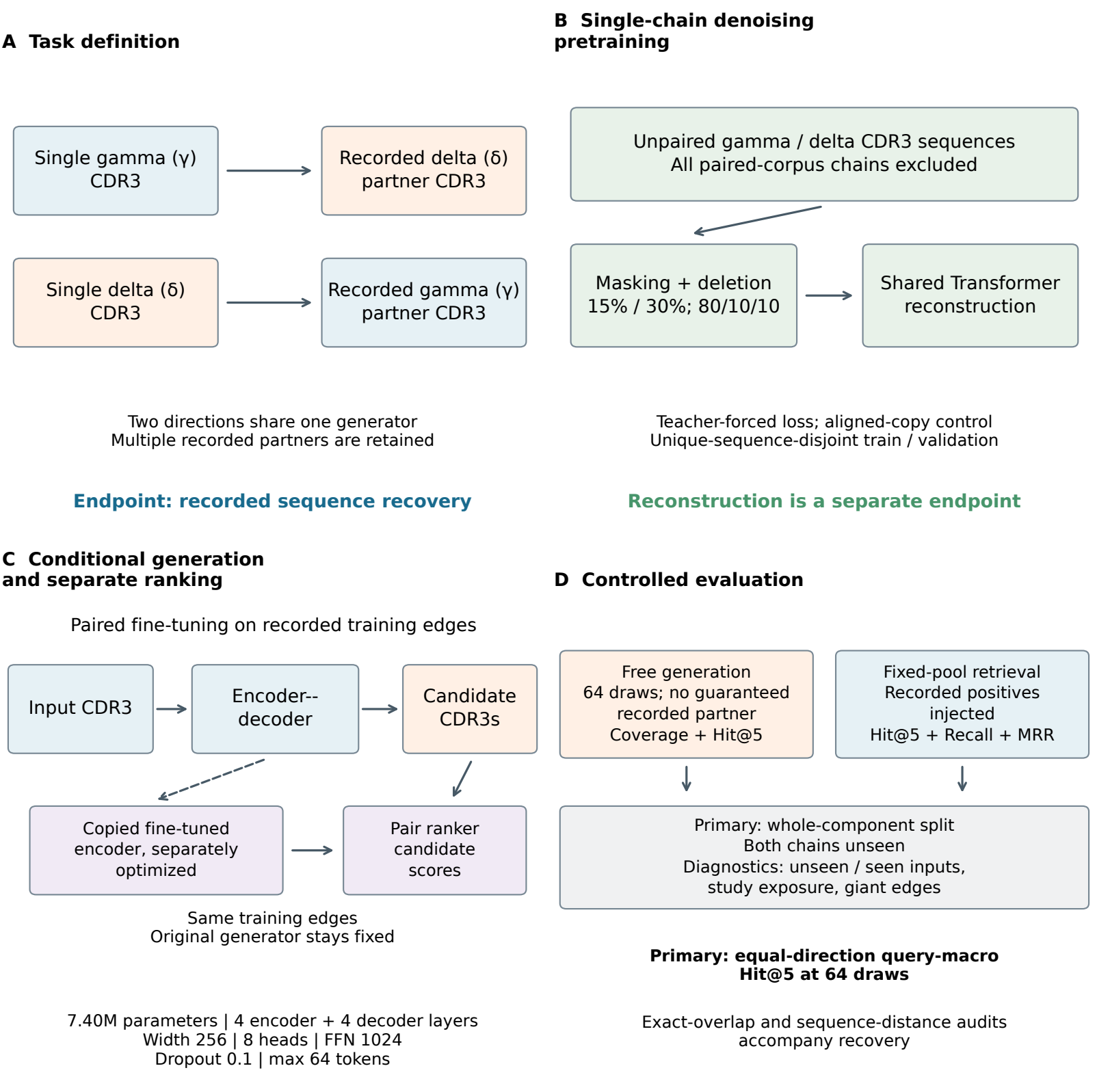
A Task definition
Single gamma (γ) CDR3
Recorded delta (δ) partner CDR3
Single delta (δ) CDR3
Recorded gamma (γ) partner CDR3
Two directions share one generator
Multiple recorded partners are retained
Endpoint: recorded sequence recovery
B Single-chain denoising pretraining
Unpaired gamma / delta CDR3 sequences
All paired-corpus chains excluded
Masking + deletion
15% / 30%; 80/10/10
Shared Transformer reconstruction
Teacher-forced loss; aligned-copy control
Unique-sequence-disjoint train / validation
Reconstruction is a separate endpoint
C Conditional generation and separate ranking
Paired fine-tuning on recorded training edges
Input CDR3
Encoder-- decoder
Candidate CDR3s
Copied fine-tuned encoder, separately optimized
Pair ranker candidate scores
Same training edges
Original generator stays fixed
7.40M parameters | 4 encoder + 4 decoder layers
Width 256 | 8 heads | FFN 1024
Dropout 0.1 | max 64 tokens
D Controlled evaluation
Free generation
64 draws; no guaranteed recorded partner
Coverage + Hit@5
Fixed-pool retrieval
Recorded positives injected
Hit@5 + Recall + MRR
Primary: whole-component split
Both chains unseen
Diagnostics: unseen / seen inputs, study exposure, giant edges
Primary: equal-direction query-macro Hit@5 at 64 draws
Exact-overlap and sequence-distance audits accompany recovery